

Integrating AI Agents into Development Workflows

Building a Code Refactoring Agent



Adam Bertram

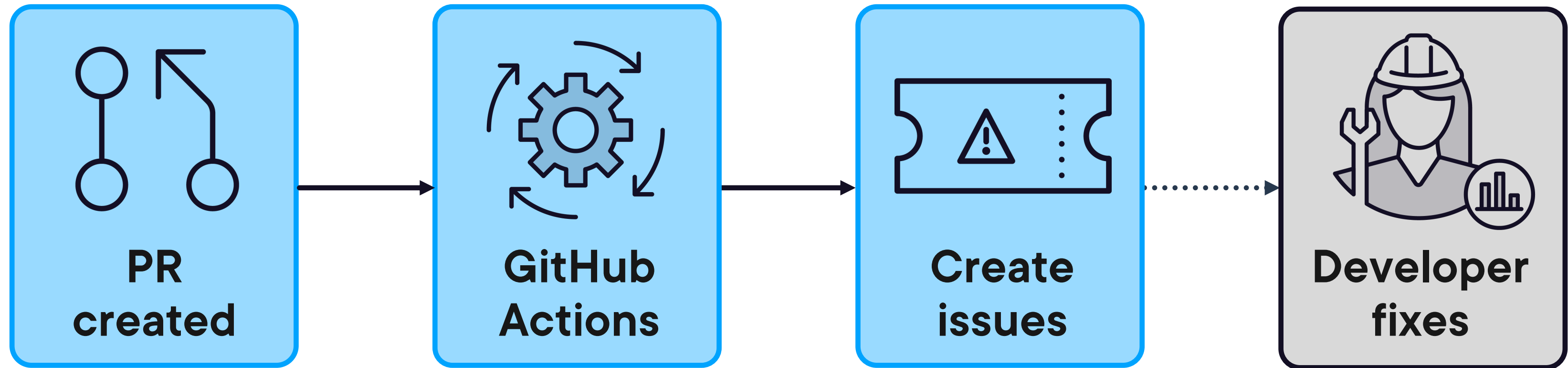
6x Microsoft MVP

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Our Current Automation Workflow

Automated



- ESLint identifies violations
- Issues labeled as "technical-debt"
- Manual interpretation of each error
- No suggested fixes provided

15% of
developer time



Showcasing the Existing Solution

Demonstrate the current GitHub workflow that detects code quality issues but requires manual intervention, showing how ESLint results become tickets that developers must manually interpret and fix.



From Automation to AI Agent



From Linting to Fixing

Developer



```
def process_order(order):  
    total = 0  
    for item in order["items"]:  
        price = item["price"]  
        quantity = item["qty"]  
        total += price * quantity  
  
    if total > 100:  
        discount = 0.1  
        total -= discount * total  
  
    return total
```

BEFORE

Developer

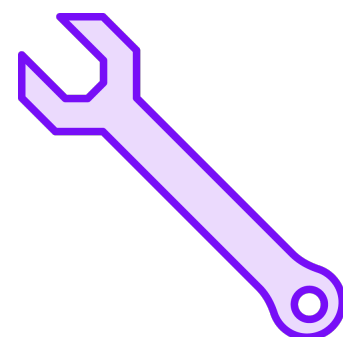


```
def process_order(order):  
    total = 0  
    for item in order["items"]:  
        price = item.get("price", 0)  
        quantity = item.get("qty", 0)  
        total += price * quantity  
  
    if total > 100:  
        discount = 0.1  
        total -= discount * total  
  
    return total
```

AFTER



Our Automation Journey

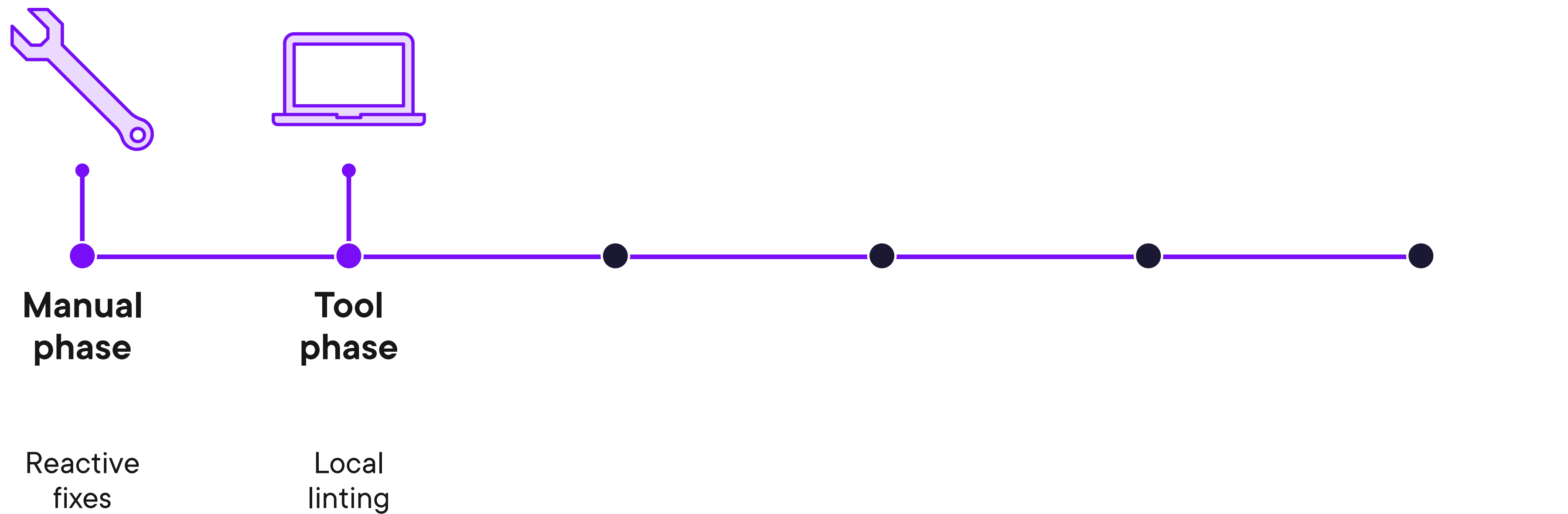


Manual
phase

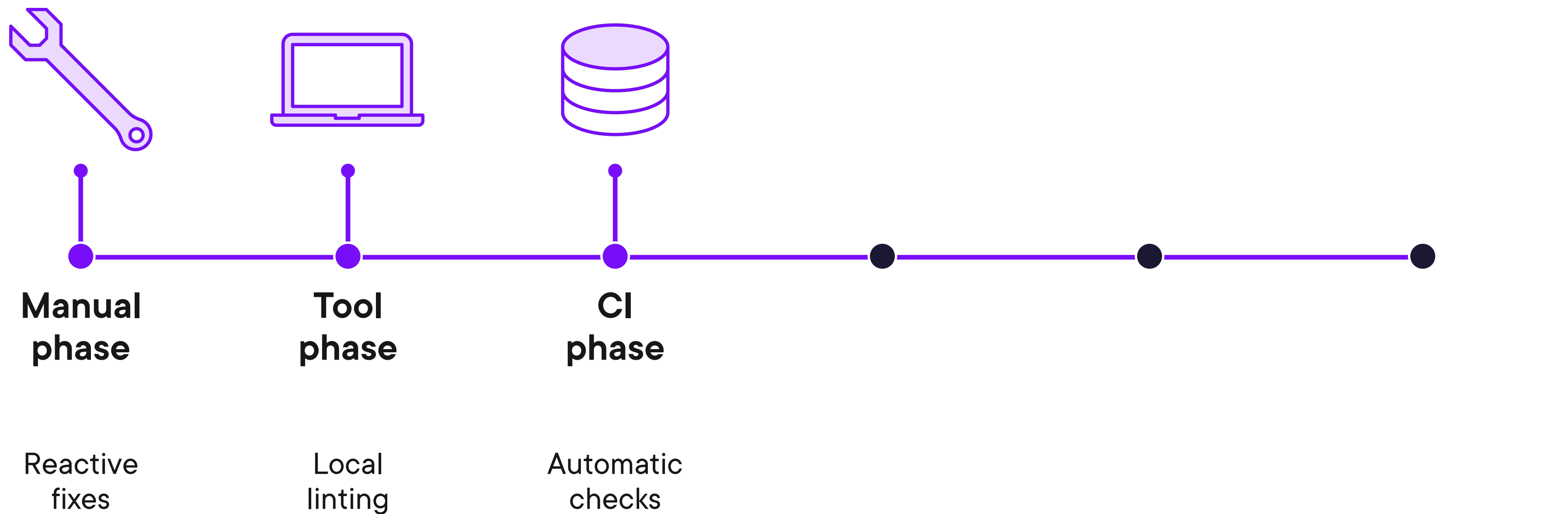
Reactive
fixes



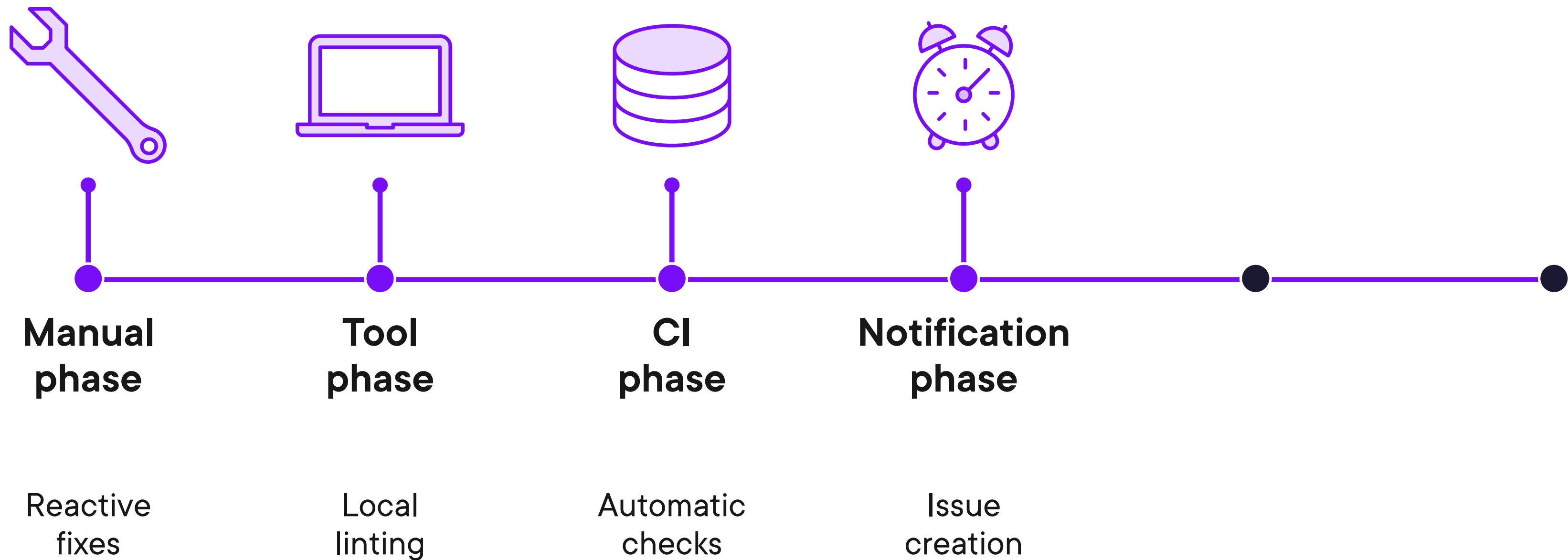
Our Automation Journey



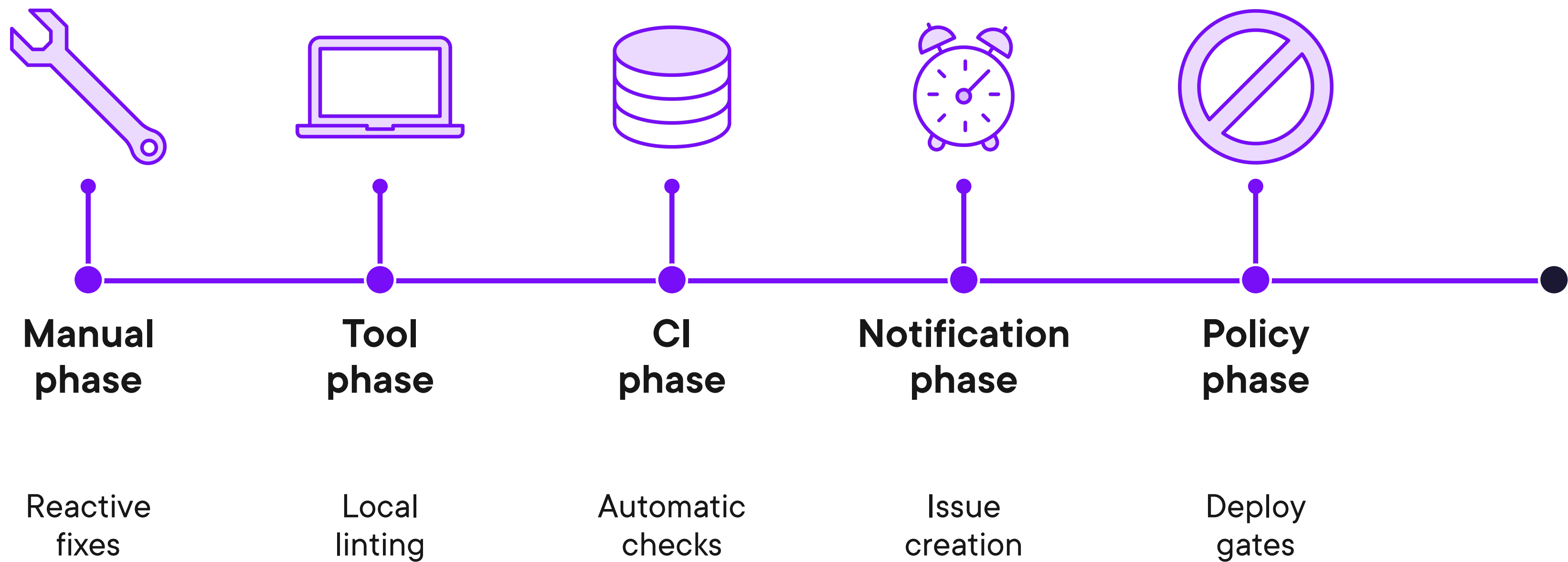
Our Automation Journey



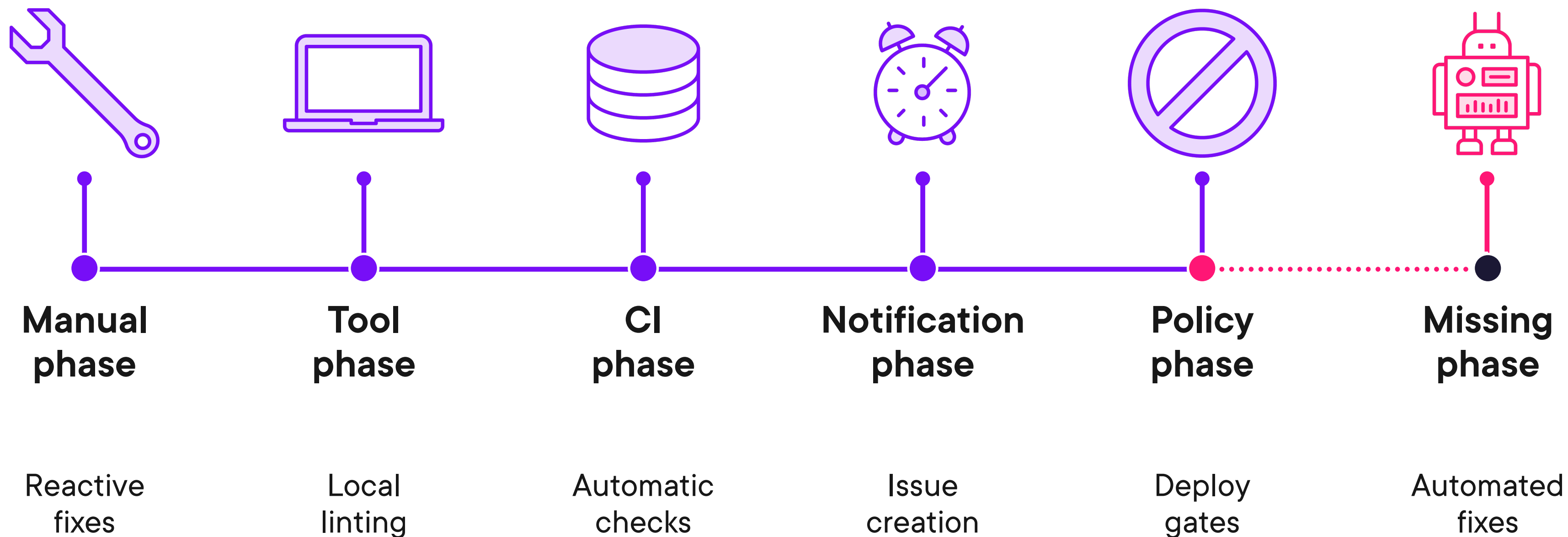
Our Automation Journey



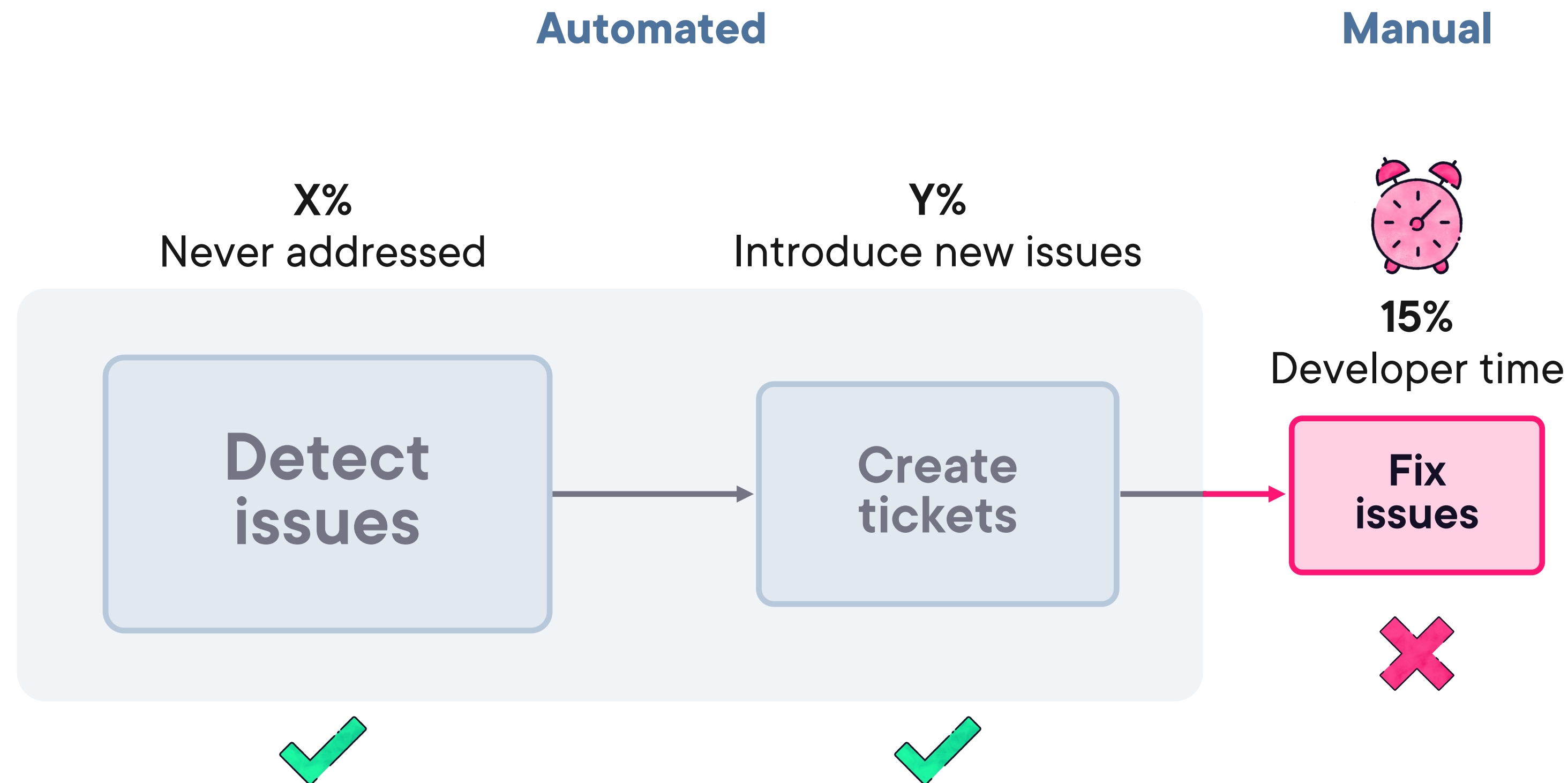
Our Automation Journey



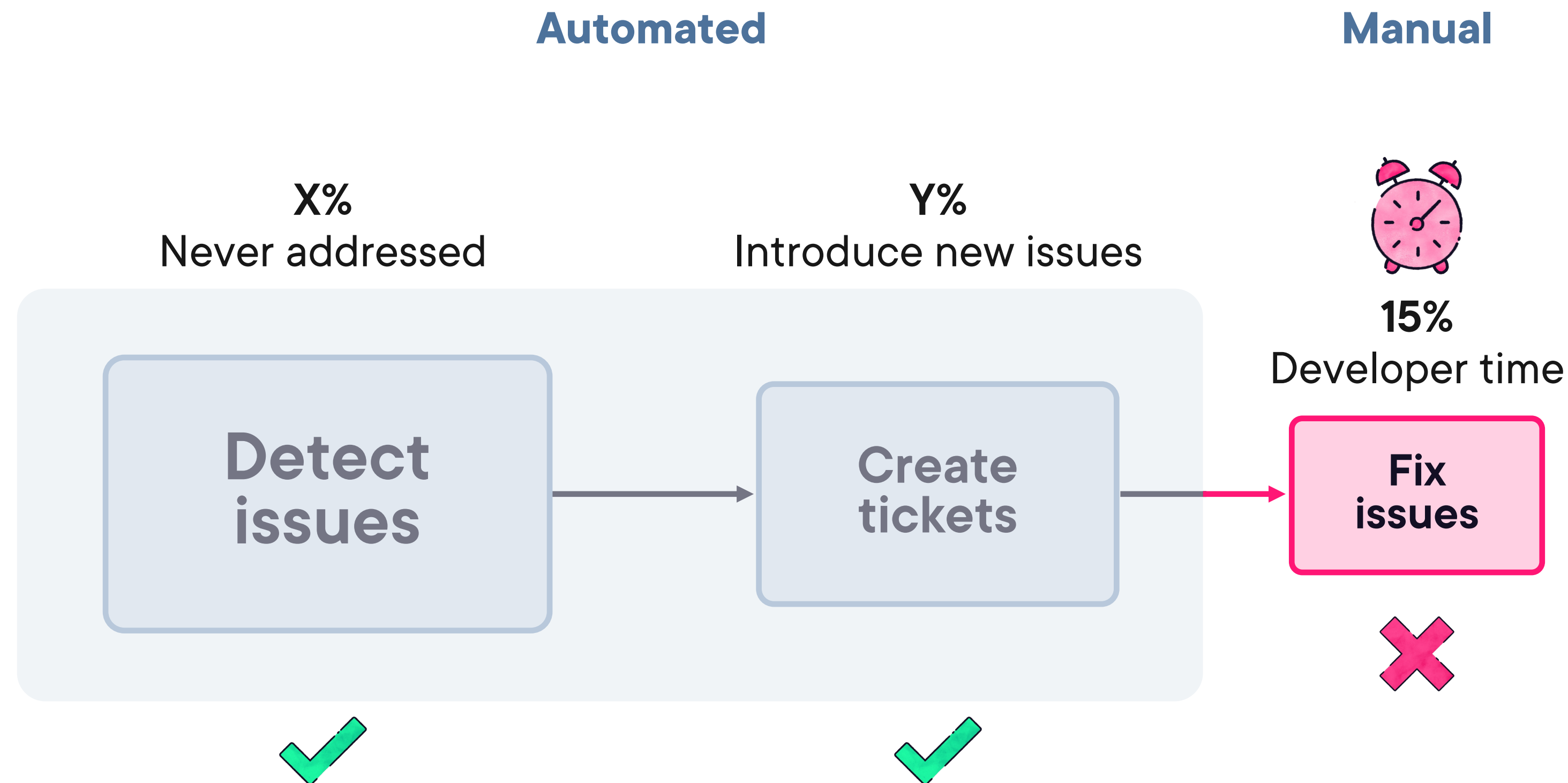
Our Automation Journey



Where Automation Falls Short

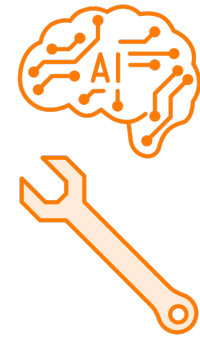


Where Automation Falls Short



AI Agents for Code Refactoring

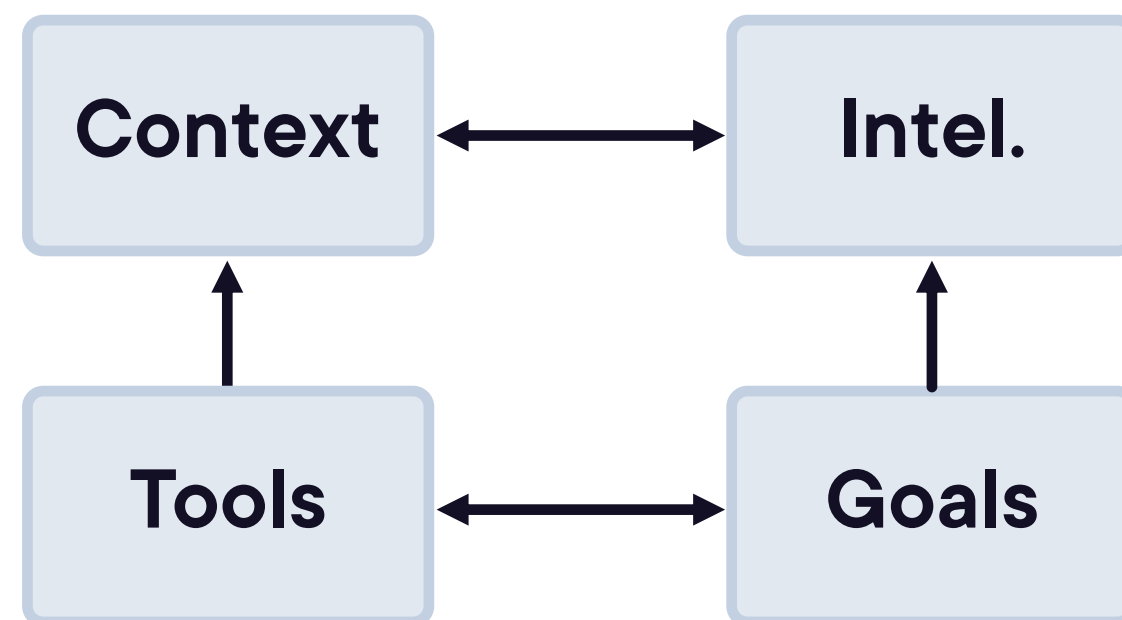
QUICK COMPARISON:



AI Agent: Complex judgment, semantics

Simple Tools: Formatting, syntax fixes

KEY COMPONENTS:



DEPLOYMENT CHECKLIST:

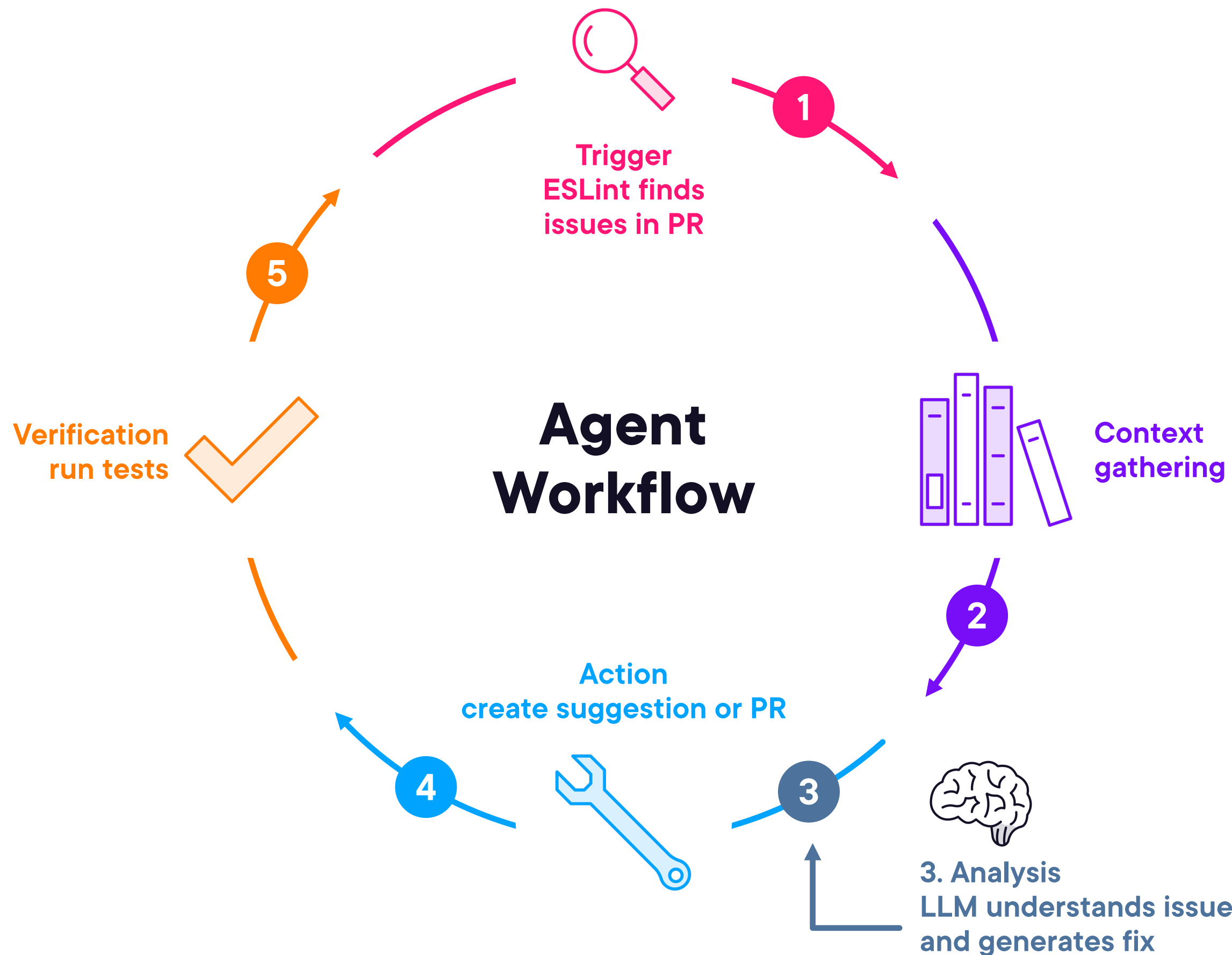
- **Pattern requires context and semantics**
- **Test verification available**
- **Safety mechanisms implemented**



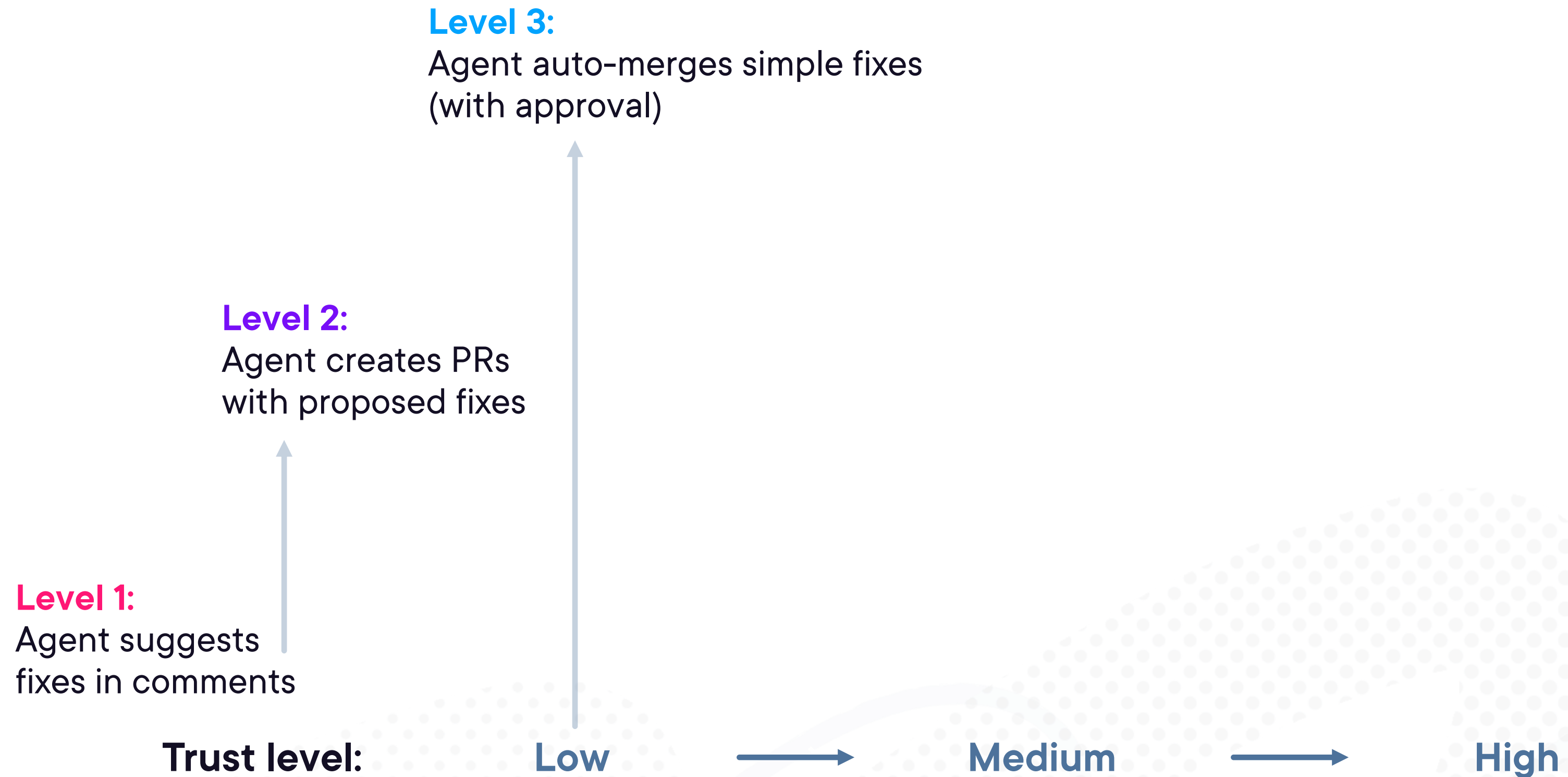
Building a “Suggesting” Agent



Building a Workflow-integrated Agent



Start Small, Scale Gradually



How Do We Know It's Working?

Issue resolution rate

85%



Developer time saved

15%



Focus on both **quality**
and **productivity**

PR acceptance
percentage

92%

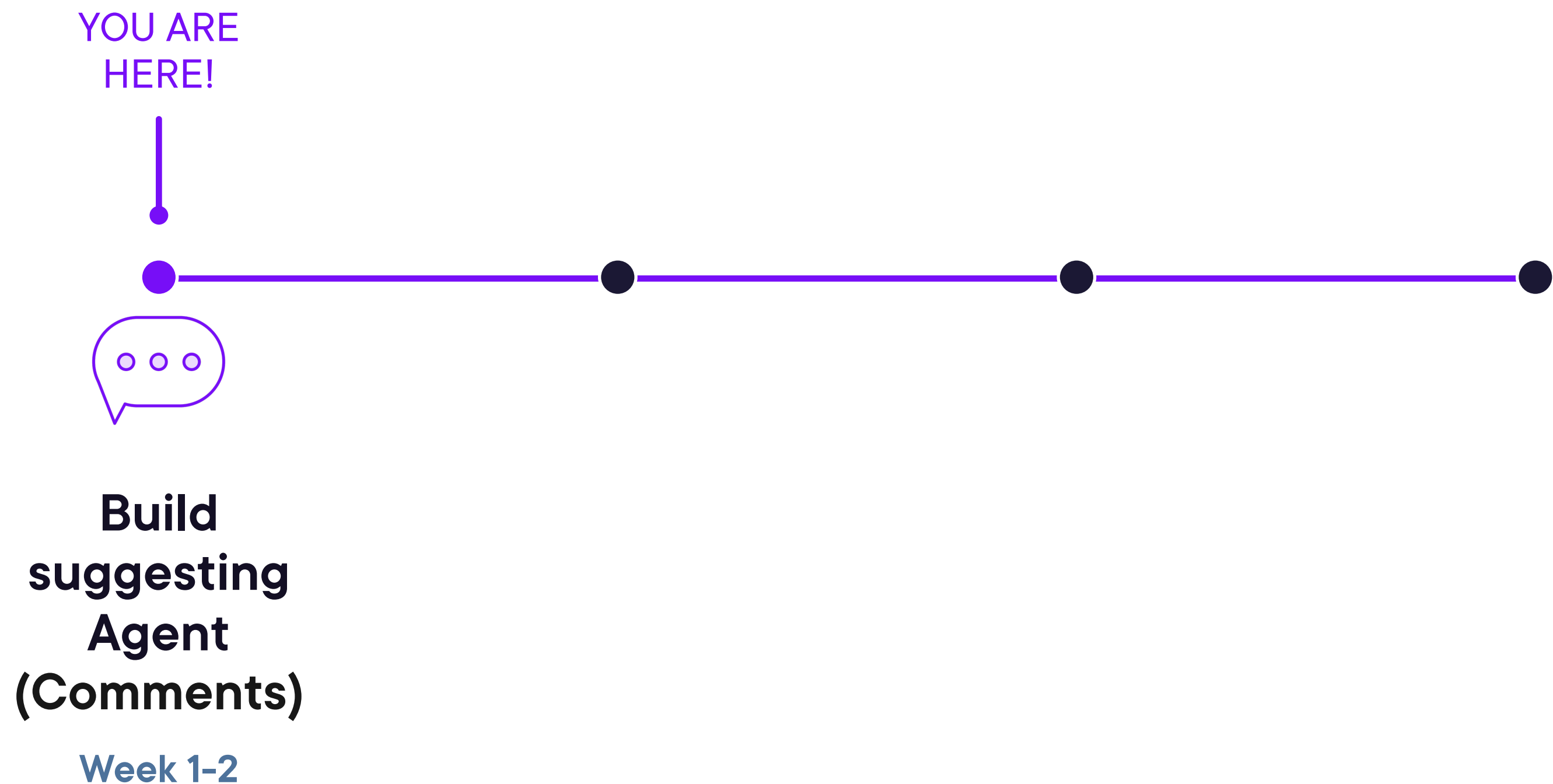


Test pass rate for
agent fixes

98%



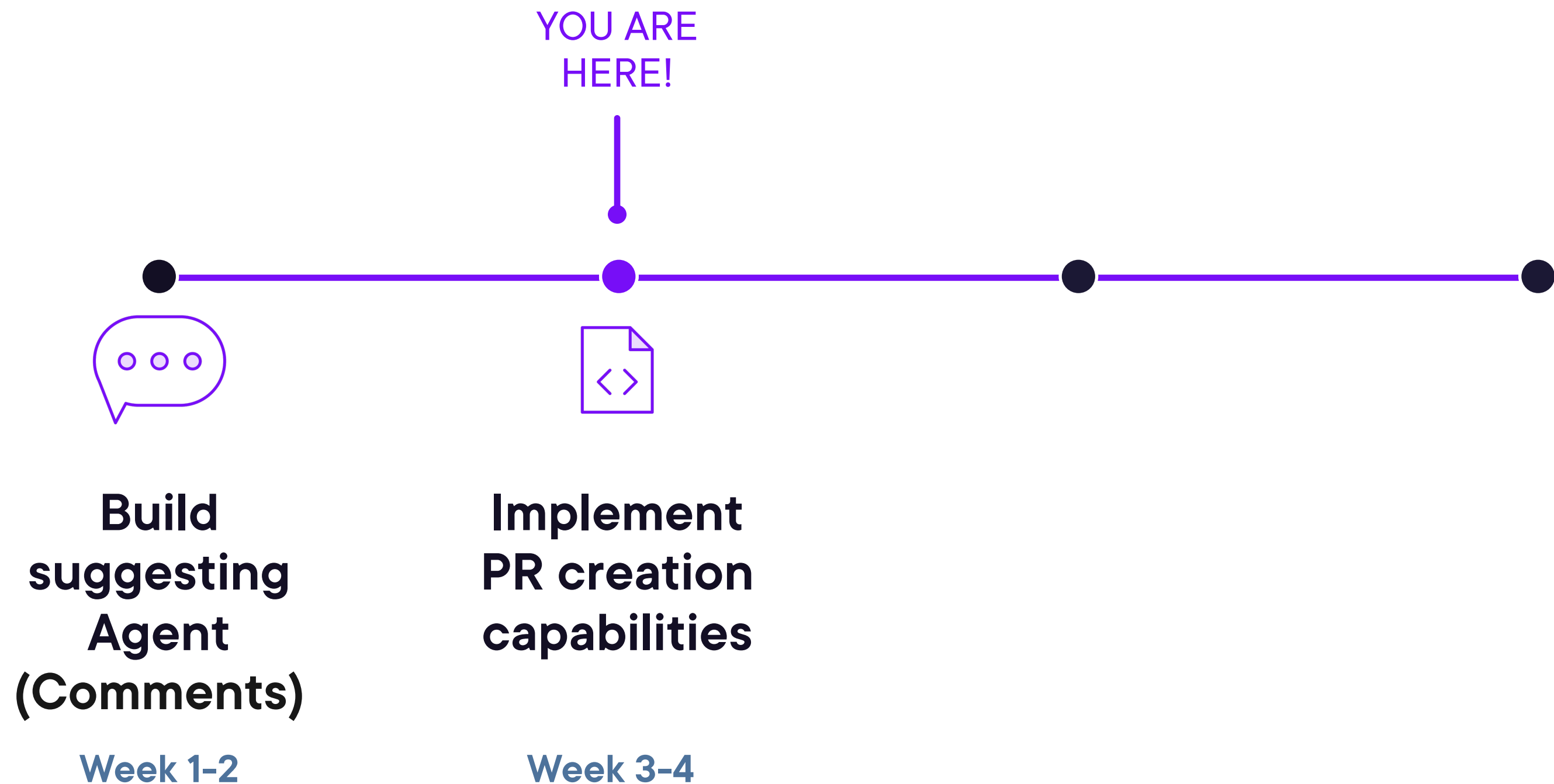
From Design to Implementation



Let's start by building the **suggesting agent**



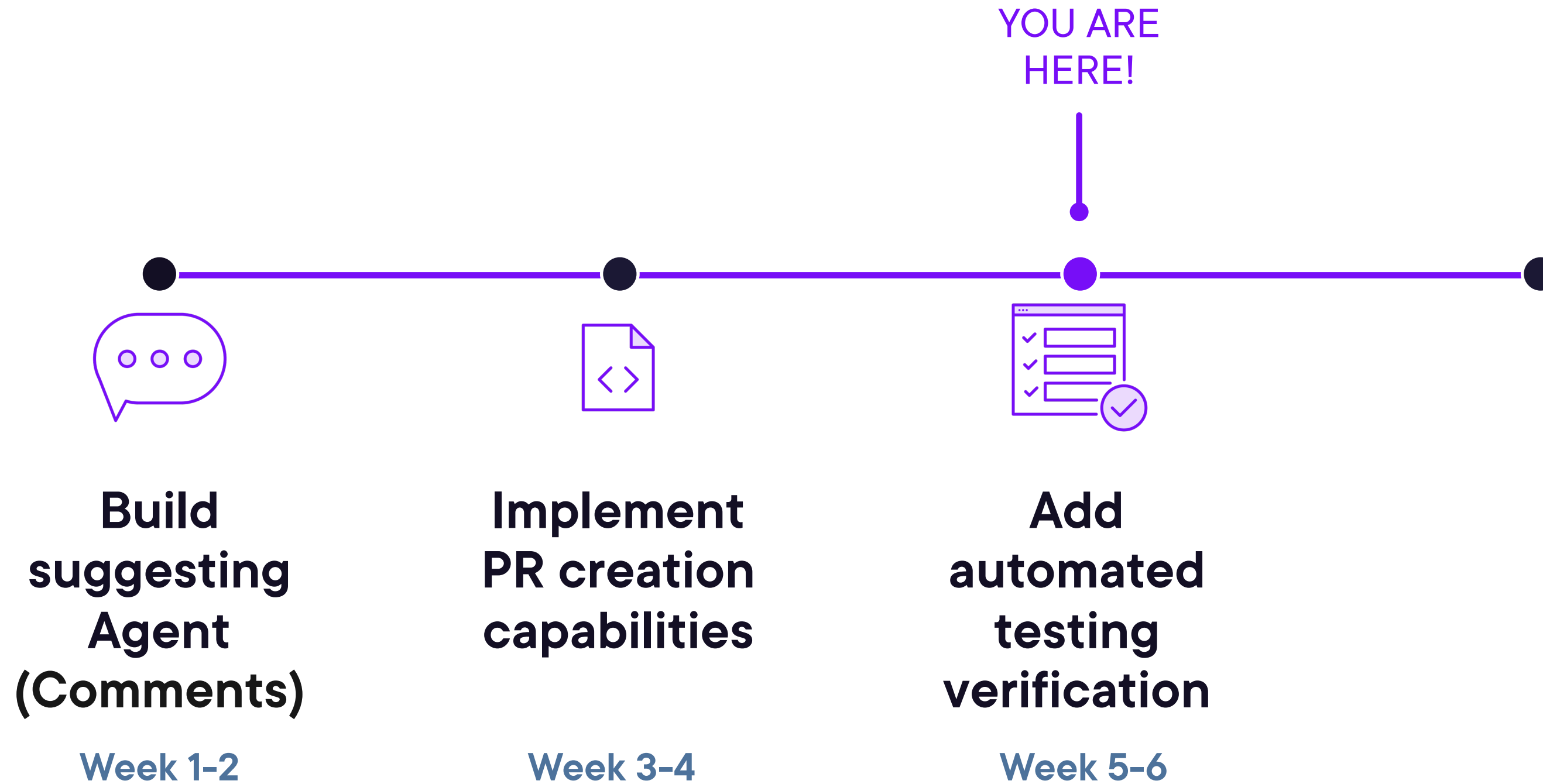
From Design to Implementation



Let's start by building the **suggesting agent**



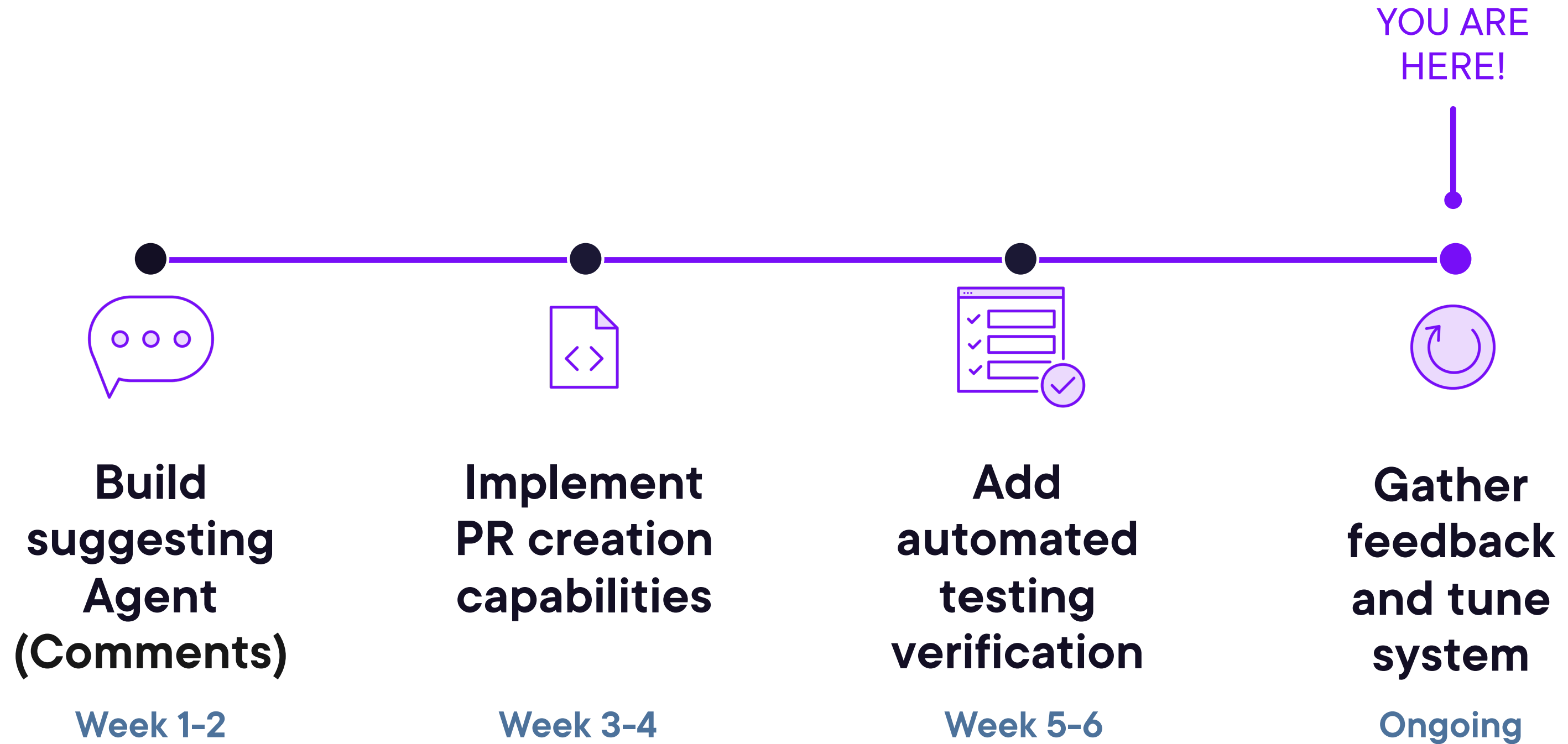
From Design to Implementation



Let's start by building the **suggesting agent**



From Design to Implementation



Let's start by building the **suggesting agent**



Demo: Implementing a “Suggesting” Agent



Implementing Automatic Pull Requests



Demo: Creating Effective Agent-generated PRs

Enhance your N8N workflow to automatically create pull requests with implemented ESLint fixes instead of just suggesting them

